WORKSHEET: REAL NUMBERS

CLASS X | SUBJECT: MATHEMATICS

SECTION A: MULTIPLE CHOICE & ASSERTION-REASON QUESTIONS (1 Mark Each)

1. Prime factors of 98 are:
(a) 2, 7 (b) 3, 7
(c) 7, 9 (d) 5, 9

2. If HCF(26, 169) = 13, then LCM(26, 169) equals:
(a) 26 (b) 52
(c) 20 (d) 338

3. If two positive integers p and q are written as $p = x^2y^2$ and $q = xy^3$, where x and y are prime numbers, then HCF(p, q) is:
(a) xy (b) xy^2
(c) x^3y^3 (d) x^2y^2

4. If "n" is a natural number, then which of the following numbers ends with zero?
(a) $(3 \times 2)^n$ (b) $(2 \times 5)^n$
(c) $(6 \times 2)^n$ (d) $(5 \times 3)^n$

5. The LCM of the smallest 2-digit number and the smallest composite number is:
(a) 12 (b) 4
(c) 20 (d) 40

6. The total number of factors of a prime number is:
(a) 1 (b) 0
(c) 2 (d) 3

7. The sum of exponents of prime factors in the prime-factorisation of 196 is:
(a) 3 (b) 4
(c) 5 (d) 2

8. If the product of two numbers a and b is 1152 and HCF(a, b) = 12, then LCM(a, b) is:
(a) 88 (b) 90
(c) 92 (d) 96

9. If LCM(x, 18) = 36 and HCF(x, 18) = 2, then x is:
(a) 2 (b) 3
(c) 4 (d) 5

10. If the prime factorisation of 2520 is $2^3 \times 3^a \times b \times 7$, then the value of $a + 2b$ is:
(a) 12 (b) 10
(c) 9 (d) 7

11. Let a and b be two positive integers such that $a = p^3q^4$ and $b = p^2q^3$, where p and q are prime numbers. If HCF(a, b) = p^mq^n and LCM(a, b) = p^rq^s, then $(m + n)(r + s)$ =
(a) 15 (b) 30
(c) 35 (d) 72

12. Let $x = a^2b^3c^n$ and $y = a^3bm^2$, where a, b, c are prime numbers. If LCM of x and y is $a^3b^4c^3$, then the value of $m + n$ is:
(a) 1 (b) 7
(c) 6 (d) 5

Directions for Q.13–17: (a) Both (A) and (R) are true and (R) is the correct explanation of (A). (b) Both (A) and (R) are true but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true.

13. **Assertion (A):** If LCM(p, q) = 30 and HCF(p, q) = 5, then $p \times q = 150$.
Reason (R): LCM(a, b) × HCF(a, b) = $a \times b$.

14. **Assertion (A):** HCF(306, 657) = 9, and LCM(306, 657) is 22338.
Reason (R): If a and b are two positive integers and HCF(a, b) = 9, then LCM(a, b) = ($a + b$)/2.

15. **Assertion (A):** 4^n ends with digit 0 for some natural number n.
Reason (R): For a number 'x' having 2 and 5 as its prime factors, x^n always ends with digit 0 for every natural number n.

16. **Assertion (A):** The HCF of two numbers is 4 and their product is 38784. Then their LCM is 9696.
Reason (R): LCM(a, b) × HCF(a, b) = $a \times b$.

17. **Assertion (A):** For any two prime numbers p and q, their HCF is 1 and LCM is ($p + q$).
Reason (R): For any two natural numbers, HCF × LCM = product of numbers.

SECTION B: SHORT ANSWER QUESTIONS (2 & 3 Marks Each)

18. Find the HCF and LCM of 72 and 120 using the prime factorisation method.

19. Express 429 as a product of its prime factors.

20. Using prime factorisation, find the HCF and LCM of 96 and 120.

21. Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number.

22. Check whether 6^n can end with the digit 0 for any natural number n.

23. Explain why $11 \times 13 \times 17 \times 19 - 17$ is a composite number.

24. Find the largest positive integer that will divide 100, 245 and 343 leaving remainders 4, 5 and 7 respectively.

25. 144 cartons of coke cans and 90 cartons of pepsi cans are to be stacked in a canteen. If each stack is of the same height and is to contain cartons of the same drink, what would be the greatest number of cartons in each stack?

26. Prove that $6 - \sqrt{7}$ is irrational, given that $\sqrt{7}$ is an irrational number.

27. Given that $\sqrt{2}$ is irrational, prove that $(5 + 3\sqrt{2})$ is an irrational number.

28. Prove that $\sqrt{2} - \sqrt{7}$ is an irrational number.

29. Prove that $\sqrt{5}$ is an irrational number.

30. Prove that $(5 - 3\sqrt{2})$ is an irrational number, given that $\sqrt{2}$ is an irrational number.

SECTION C: LONG ANSWER QUESTIONS (5 Marks Each)

31. Amit, Sunita and Sumit start preparing cards for all the persons in an old age home. In order to complete one card, they take 10, 16 and 20 minutes respectively. If they all started together, after what time will they begin preparing a new card together?

32. Aakriti decided to distribute milk in an orphanage on her birthday. The supplier brought two milk containers which contain 398 litres and 436 litres of milk. The milk is to be transferred to another vessel using a drum so that 7 litres and 11 litres of milk is left in both the containers respectively. What will be the maximum capacity of the drum?

33. Find the greatest number that will divide 382, 509 and 636 leaving remainders 4, 5 and 6 respectively.

SECTION D: CASE STUDY BASED QUESTIONS (4 Marks Each)

34. Teaching Mathematics through activities is a powerful approach that enhances student's understanding and engagement. Keeping this in mind, Ms. Mukta planned a prime number game for class 5 students. She announced the number 2 in her class and asked the first student to multiply it by a prime number and then pass it to second student. Second student also multiplied it by a prime number and passed it to third student. In this way by multiplying with a prime number, the last student got 173250.

Based on the above information, answer the following questions:

(i) What is the least prime number used by students?

(ii) Which prime number has been used maximum times?

(iii) (a) How many students are in the class? **OR** (b) What is the highest prime number used by students?

35. Khushi wants to organise her birthday party. She is very health conscious, thus she decided to serve only fruits at the party. She has 36 apples and 60 bananas at home and decided to serve them. She does not want to distribute the fruits equally among all.

Based on the above information, answer the following questions:

(i) How many guests Khushi can invite at the most?

(ii) How many apples and bananas will each guest get?

(iii) (a) If Khushi decides to add 42 mangoes, how many maximum guests can she invite? **OR** (b) If she decides to remove 6 apples, how many maximum guests can she invite?

36. On the 71st Republic Day Parade in Delhi, Captain RS Meel is organizing the parade for the following two groups:
(a) The first group consists of an Army contingent with 624 members, marching behind an army band of 32 members.
(b) The second group comprises CRPF troops with 468 soldiers, marching behind 228 bikers. Both groups are to march in the same number of columns. Following the sequence of soldiers, various state tableaux will showcase the culture of their respective states.

Based on the above information, answer the following questions:

(i) What is the maximum number of columns in which the army troops can march?

(ii) What is the maximum number of columns in which the CRPF troops can march?

(iii) (a) What minimum number should be subtracted from the number of CRPF soldiers and the number of bikers so that their maximum number of columns is equal to the maximum number of columns of army troops? **OR** (b) What minimum number should be added with the number of CRPF soldiers and the number of bikers so that their maximum number of columns is equal to the maximum number of columns of army troops?